

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Total Marks:20

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I K.S.Dharshana(192412015) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well- explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

ANSWERS FOR ABOVE QUESTIONS:

1.A Smart home automation system uses AI to automatically control lights, temperature and security based on environmental conditions. The system receive input from sensors and make intelligent decisions to improve comfort, energy efficiency and safety.

Types of Intelligent Agents

1. Simple Reflex Agent

A Simple Reflex Agent makes decisions only by looking at the current situation. It does not remember what happened earlier.

Example in a Smart Home:

- **Turns on the light when motion is detected.**
- **Turns off the light when no one is in the room.**

Advantages

- **Responds quickly.**
- **Easy to design and implement.**

Disadvantages

- **Cannot learn user habits.**
 - **Cannot make decisions using past information.**
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2. Model-Based Agent

A Model-Based Agent keeps track of previous information and maintains an internal model of the environment. This helps it make better decisions.

Example in a Smart Home:

- **Remembers that the living room is usually occupied in the evening.**
- **Detects if a window has been left open before turning on the AC.**
- **Uses previous sensor readings to predict user activity.**

Advantages

- **Works well even when all information is not available.**
- **Makes smarter decisions by using memory.**

Disadvantages

- **More complex than a Simple Reflex Agent.**
 - **Requires additional memory and processing.**
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3. Goal-Based Agent

A Goal-Based Agent works to achieve a specific goal. Before taking an action, it checks which option will help it reach that goal.

Example in a Smart Home:

The goals may include:

- **Maintaining a comfortable room temperature.**
- **Saving electricity.**
- **Keeping the house secure.**

Advantages

- **Plans its actions carefully.**
- **Focuses on achieving the desired goal.**

Disadvantages

- **Needs more processing time.**
 - **Depends on clearly defined goals.**
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4. Utility-Based Agent

A Utility-Based Agent selects the action that gives the best overall result. Instead of focusing on only one goal, it considers several factors like comfort, energy saving, and security before making a decision.

Example in a Smart Home:

If the room becomes warm, instead of setting the AC to the lowest temperature, it chooses a comfortable temperature that also saves electricity.

Advantages

- **Makes balanced and efficient decisions.**
- **Considers multiple factors at the same time.**

Disadvantages

- **Utility calculations can be complicated.**
 - **Needs more computational power.**
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Best Approach

A combination of Model-Based Agent and Utility-Based Agent is the best choice for a smart home system.

This combination allows the system to remember user habits, understand the current environment, and make decisions that provide maximum comfort while reducing electricity usage and maintaining security. For example, if the user usually reaches home at 6 PM, the system can automatically cool the room before arrival while keeping unnecessary appliances switched off.

Conclusion

Although each intelligent agent has its own strengths, a hybrid approach using Model-Based and Utility-Based Agents is the most suitable for a smart home because it can learn user preferences, remember previous activities, and make intelligent decisions that improve comfort, energy efficiency, and security.

2. Robot Navigation in an Unknown Maze Using Search Algorithms (10 Marks)

Introduction

Imagine a robot placed inside a maze without knowing its layout. Its task is to find the exit by exploring different paths. Since the robot has no prior knowledge of the maze, AI search algorithms help it decide which path to take. Two commonly used uninformed search algorithms are Uniform Cost Search (UCS) and Iterative Deepening Search (IDS).

Uniform Cost Search (UCS)

Uniform Cost Search always chooses the path with the lowest total cost. Instead of selecting the shortest path based on the number of steps, it considers the actual cost of moving through each path.

How it works

- The robot starts from its initial position.
- It calculates the cost of every possible path.
- It always explores the path with the smallest cost first.
- The process continues until the robot reaches the exit.

Advantages

- Finds the lowest-cost path.
- Guarantees an optimal solution.
- Works well when different paths have different movement costs.

Disadvantages

- Requires more memory.
 - Can be slower in large mazes.
 - Time Complexity: $O(b^{(1 + C^*/\epsilon)})$
 - Space Complexity: $O(b^{(1 + C^*/\epsilon)})$
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Iterative Deepening Search (IDS)

Iterative Deepening Search combines the advantages of Breadth-First Search and Depth-First Search. It searches one level at a time and gradually increases the search depth.

How it works

- First searches at depth 0.
- If the exit is not found, searches at depth 1.
- Continues increasing the depth until the exit is located.

Advantages

- **Uses very little memory.**
- **Guarantees finding the solution if one exists.**
- **Suitable when the depth of the goal is unknown.**

Disadvantages

- **Repeats the search at every depth.**
- **May take extra time because of repeated exploration.**
- **Time Complexity: $O(b^d)$**
- **Space Complexity: $O(bd)$**

Relation to Intelligent Agents

A Simple Reflex Agent only reacts to the current situation, so it may repeatedly visit the same locations and is not suitable for maze navigation.

A Model-Based Agent remembers previously visited paths and avoids unnecessary repetition, making exploration more efficient.

A Goal-Based Agent focuses on reaching the exit. It uses search algorithms to choose the best path towards the goal.

A Utility-Based Agent considers factors such as distance, energy consumption, and safety before deciding which path to follow.

Best Combination

The most suitable solution is a Goal-Based Agent with Uniform Cost Search (UCS).

The Goal-Based Agent keeps the robot focused on reaching the exit, while UCS ensures that the path chosen has the lowest possible cost. This combination provides an efficient and optimal solution, especially when different paths require different amounts of energy or effort.

If the robot has limited memory, a Model-Based Agent with Iterative Deepening Search (IDS) is another good choice because it remembers explored paths while using less memory.

Conclusion

Both UCS and IDS are useful for solving maze problems. UCS is best when finding the lowest-cost path is important, whereas IDS is useful when memory is limited and the depth of the goal is unknown. Overall, using a Goal-Based Agent with Uniform Cost Search provides the most effective solution for intelligent maze navigation.